

IMO(16th) – 1974

First Day

1. Alice, Betty and Carol took the same series of examinations. There was one grade of A, one grade of B and one grade of C for each examination, where A, B, C are different positive integers. The final test scores were Alice – 20, Betty – 10, Carol – 9. If Betty placed first in the arithmetic examinations, who placed second in the spelling examination?
2. Let $\triangle ABC$ be a triangle. Prove that there exists a point D on the side AB such that CD is the geometric mean of AD and BD if and only if $\sqrt{\sin A \sin B} \leq \sin \frac{C}{2}$.
3. Prove that there does not exist a natural number n for which the number $\sum_{k=0}^n \binom{2n+1}{2k+1} 2^{3k}$ is divisible by 5.

Second Day

4. Consider a partition of 8×8 chessboard into p rectangles whose interiors are disjoint such that each rectangle contains an equal number of white and black cells. Assume that $a_1 < a_2 < \dots < a_p$ where a_i denotes the number of white cells in the ith rectangle. Find the maximal p for which such a partition is possible and for that p determine all possible corresponding sequences $a_1, a_2, \dots, a_p$.
5. If a, b, c, d are arbitrary positive real numbers, find all possible values of :

$$S = \frac{a}{a+b+d} + \frac{b}{a+b+c} + \frac{c}{b+c+d} + \frac{d}{a+c+d}$$
6. Let $P(x)$ be a polynomial with integer coefficients. If $n(P)$ is the number of (distinct) integers k such that $P^2(k) = 1$, prove that $n(P) - \deg(P) \leq 2$, where $\deg(P)$ denotes the degree of the polynomial P.

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